

# Uncertainty-Calibrated Dual-Path Adaptation for Test-Time Adaptation

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## Abstract

Test-Time Adaptation (TTA) aims to adapt a source-pretrained model to a shifting target distribution using only unlabeled online data. While the recently proposed COME method advances this setting by incorporating evidential (Dirichlet) uncertainty into a conservative opinion-entropy objective, it treats every test sample uniformly: the same conservative entropy loss is applied regardless of whether the model is genuinely uncertain or already confident. This uniform treatment discards useful supervisory signal that confident samples could otherwise provide and offers no mechanism linking the evidential uncertainty to the softmax confidence used for prediction. We propose **Uncertainty-Calibrated Dual-Path Adaptation (UCDPA)**, a framework that resolves these limitations through three components. First, we introduce the *Uncertainty-Confidence Balance (UCB) score*, a differentiable routing signal  $b_i = \sigma((c_i - \tau_c)/T_c) \sigma((1 - u_i - \tau_u)/T_u)$  that is monotonically increasing in softmax confidence  $c_i$  and monotonically decreasing in Dirichlet uncertainty  $u_i$ , enabling per-sample soft routing between two complementary objectives. Second, we design a *scale-normalized dual-path* adaptation loss that applies conservative opinion-entropy minimization to uncertain samples and pseudo-label cross-entropy to confident samples, with class-balanced reweighting and a calibration regularizer that maintains batch-mean consistency between confidence and uncertainty. Third, we add a feature prototype alignment term that pulls representations toward their pseudo-label class prototypes. We provide a theorem on UCB monotonicity and a proposition on calibration consistency. On ImageNet-C with ViT-B/16, UCDPA improves mean corruption accuracy by +3.07 percentage points over COME (69.20 vs. 66.13), with statistically significant gains ( $p < 0.001$ ), lower Expected and Maximum Calibration Error (3.48 vs. 7.22), and improved error-detection AUROC (85.22 vs. 84.34). We further validate UCDPA on CIFAR-100-C with a CIFAR-ResNet-18 backbone (BatchNorm adaptation), demonstrating cross-backbone and cross-dataset generalization. Ablations confirm that every component contributes, and sensitivity sweeps show the method is stable to routing hyperparameter choices within a 2.54 pp range.

## 1 Introduction

Modern vision models are trained on a fixed source distribution and deployed on a target distribution that inevitably drifts. Even strong ImageNet-pretrained transformers suffer

large accuracy drops when the test data is corrupted by noise, blur, weather, or digital artifacts (Dosovitskiy et al., 2021; Hendrycks and Dietterich, 2019). *Test-Time Adaptation (TTA)* addresses this gap by adapting the model using only unlabeled data observed at test time, without access to the original source data or any labels (Schneider et al., 2020; Sun et al., 2020; Wang et al., 2021). Because the target stream is unlabeled and non-stationary, TTA methods must extract a usable learning signal from the model’s own predictions while simultaneously avoiding catastrophic failure modes such as confirmation bias, class collapse, and over-confidence.

A recent line of work, culminating in COME (Conservative Opinion-Minimization Entropy) (Author et al., 2024), advances TTA by grounding the adaptation objective in *evidential deep learning*. COME places a Dirichlet distribution over the class-probability simplex, derives a per-sample *uncertainty mass*  $u_i = K/S_i$  from the Dirichlet strength  $S_i$ , and combines the standard softmax entropy with an *opinion entropy* defined over the belief masses plus the uncertainty mass. The resulting conservative entropy loss down-weights the contribution of uncertain samples, providing a degree of robustness that pure entropy minimization (Wang et al., 2021) lacks. Empirically, COME improves over Tent (Wang et al., 2021), EATA (Niu et al., 2022), and SAR (Niu et al., 2023) on standard corruption benchmarks.

Despite its successes, COME has three structural limitations that motivate our work.

**Limitation 1: Uniform treatment of samples.** COME applies the *same* conservative entropy objective to every test sample, differing only in the magnitude of the per-sample gradient through the (implicit) entropy weighting. There is no mechanism to route genuinely confident samples to a stronger supervisory signal. When the model is already confident and correct on a sample, conservative entropy provides a vanishing gradient, wasting a sample that could otherwise reinforce the decision boundary through a sharper pseudo-label cross-entropy objective.

**Limitation 2: No pseudo-label leverage.** Pseudo-labeling has driven large gains in semi-supervised learning (Lee, 2013; Sohn et al., 2020; Xie et al., 2020) and has been shown to provide stronger gradients than entropy minimization when the model is confident (Chen et al., 2023; Prabhu et al., 2023a). COME does not leverage pseudo-labels at all, leaving a known source of signal on the table. A natural question is whether one can *combine* the safety of conservative entropy (for uncertain

samples) with the strength of pseudo-label cross-entropy (for confident samples) within a single TTA framework.

**Limitation 3: No calibration guarantee.** COME uses evidential uncertainty to reweight the loss but never enforces any agreement between the evidential uncertainty  $u_i$  and the softmax confidence  $c_i = \max_k p_{ik}$  that actually drives predictions. The two quantities can drift arbitrarily far apart during adaptation: the model can become highly confident in its softmax output while the Dirichlet uncertainty remains large, or vice versa. This decoupling undermines both the reliability of the uncertainty estimate (used for error detection) and the trustworthiness of the confidence estimate (used for routing and selective prediction).

**Our insight.** The two quantities that COME computes—softmax confidence  $c_i$  and Dirichlet uncertainty  $u_i$ —are complementary measures of the model’s epistemic state. *Confidence* captures the sharpness of the predictive distribution, while *uncertainty* captures the total evidence mass. A sample that is *both* confident ( $c_i \rightarrow 1$ ) and certain ( $u_i \rightarrow 0$ ) is a reliable candidate for pseudo-label supervision; a sample that is uncertain or low-confidence should be handled conservatively. The key realization is that these two measures should be *balanced* to route samples, and that this balance should be *calibrated* so the two views of the model’s epistemic state remain consistent.

**Our method: UCDPA.** We propose **Uncertainty-Calibrated Dual-Path Adaptation (UCDPA)**, which operationalizes this insight through four components:

1. A novel **UCB (Uncertainty-Confidence Balance) score**  $b_i = \sigma((c_i - \tau_c)/T_c) \sigma((1 - u_i - \tau_u)/T_u)$  that quantifies, per sample, the degree to which the model is *both* confident and certain. We prove (Theorem 1) that  $b_i$  is monotonically increasing in  $c_i$ , monotonically decreasing in  $u_i$ , and achieves its maximum only when  $c_i \rightarrow 1$  and  $u_i \rightarrow 0$ .
2. A **scale-normalized dual-path adaptation loss** that softly routes each sample between conservative opinion-entropy minimization (Path 1, for uncertain/low-confidence samples) and pseudo-label cross-entropy (Path 2, for confident/certain samples), with per-batch scale normalization ensuring both paths contribute comparable gradient magnitudes, weighted by a class-balanced inverse-frequency term from an exponential moving average (EMA) class tracker.
3. A **calibration regularizer**  $L_{\text{cal}} = |\mathbb{E}[c] - \mathbb{E}[1 - u]|$  that explicitly binds the batch-mean softmax confidence and the batch-mean evidential uncertainty. We show (Proposition 1) that minimizing this regularizer bounds the mean discrepancy between the two uncertainty views, with residual per-sample discrepancy controlled by their joint variance.
4. A **feature prototype alignment** term that maintains an EMA prototype bank and pulls features toward their

pseudo-label class prototype, improving representation coherence under shift.

**Contributions.** Our contributions are:

1. A **novel UCB score** that quantifies the balance between Dirichlet uncertainty and softmax confidence for per-sample routing, with a formal monotonicity theorem (Theorem 1) establishing that the routing probability is monotonically increasing in confidence and monotonically decreasing in uncertainty.
2. A **dual-path adaptation framework** that leverages both conservative entropy minimization (for uncertain samples) and pseudo-label cross-entropy (for confident samples), combined with class-balanced reweighting that prevents class collapse and a calibration regularizer that maintains batch-mean consistency between the two uncertainty views (Proposition 1).
3. **Comprehensive experiments** on ImageNet-C with ViT-B/16 showing a +3.07 percentage-point accuracy improvement over the COME baseline (69.20 vs. 66.13), with statistical significance ( $p < 0.001$ ), improved calibration (ECE 3.48 vs. 7.22; MCE 7.53 vs. 12.21), and improved uncertainty quantification (Predictive Entropy, Mutual Information) and error detection (AUROC 85.22 vs. 84.34). We compare with six recent baselines including CoTTA, MEMO, POEM, and FOA, showing UCDPA outperforms all by +3.51–11.42 pp. We also validate on CIFAR-100-C with a CIFAR-ResNet-18 backbone (cross-backbone, cross-dataset), and provide sensitivity sweeps (4 routing hyperparameters), batch-size experiments ( $B \in \{1, 8, 32, 64\}$ , including  $B = 1$  TTA), a scale-normalization ablation, and a theoretical analysis of routing stability, error amplification, and convergence.

The remainder of the paper is organized as follows. Section 2 surveys related work. Section 3 develops the UCDPA method, including the UCB score, dual-path loss, calibration regularizer, prototype alignment, and the monotonicity theorem and consistency proposition. Section 4 reports experiments on ImageNet-C. Section 5 discusses why the method works, its limitations, and threats to validity. Section 6 concludes.

## 2 Related Work

### 2.1 Test-Time Adaptation

Test-Time Adaptation adapts a source-pretrained model to a shifting target distribution using only unlabeled test data. *Normalization-based* methods (Nado et al., 2020; Schneider et al., 2020) recompute batch statistics—efficient but limited in expressiveness. *Entropy-based* methods minimize softmax entropy: Tent (Wang et al., 2021) updates normalization affine parameters, but is prone to confirmation bias (Mummadi et al., 2021; Niu et al., 2022); EATA (Niu et al., 2022) adds reliable sample weighting and Fisher regularization; SAR (Niu et al.,

2023) uses Sharpness-Aware Minimization. *Augmentation-based* methods include TTT (Sun et al., 2020), MEMO (Zhang et al., 2022), and CoTTA (Wang et al., 2022) (teacher-student with stochastic restore). For ViT (Dosovitskiy et al., 2021), LayerNorm-only adaptation is standard (Author et al., 2024; Niu et al., 2023).

## 2.2 Uncertainty, Calibration, and Pseudo-Labeling

UCDPA relies on *evidential deep learning* (EDL) (Amini et al., 2020; Sensoy et al., 2018), which places a Dirichlet distribution over class probabilities in a single forward pass, yielding belief masses and an uncertainty mass. Unlike Bayesian neural networks (Abdar et al., 2021; Kendall and Gal, 2017) and MC Dropout (Gal and Ghahramani, 2016), EDL is suitable for online TTA. COME (Author et al., 2024) adopts EDL inside TTA, which we extend. Post-hoc calibration methods (temperature scaling (Guo et al., 2017), Platt scaling (Platt, 1999)) require labeled sets unavailable in TTA; our regularizer operates during adaptation without labels (Proposition 1). Pseudo-labeling underpins SSL (Lee, 2013; Sohn et al., 2020; Xie et al., 2020) and TTA (Chen et al., 2023; Prabhu et al., 2023a), but risks confirmation bias. Selective pseudo-labeling via confidence thresholds is well-established (Niu et al., 2022); UCDPA’s UCB routing extends this by using a *product* of two sigmoids (confidence *and* evidential certainty) and routing *softly* rather than filtering hard. The product form is critical: a confidently-wrong sample (high  $c$ , high  $u$ ) passes a confidence-only filter but is correctly routed to the conservative path by the UCB product.

## 2.3 COME and Recent TTA Paradigms

COME (Author et al., 2024) is our primary baseline. It computes evidential quantities from logits via  $p$ -norm normalization, forms Dirichlet evidence, and defines an opinion entropy. The conservative entropy loss blends softmax and opinion entropy. UCDPA retains COME’s evidential machinery as Path 1, but addresses its three limitations: (i) we route samples rather than treating them uniformly, (ii) we add a pseudo-label cross-entropy path, and (iii) we add a calibration regularizer (Proposition 1).

**Comparison with CoTTA, MEMO, POEM, FOA, and STAMP.** We compare empirically with CoTTA (Wang et al., 2022), MEMO (Zhang et al., 2022), POEM (Yi et al., 2025), and FOA (Niu et al., 2024) on ImageNet-C/ViT-B/16 (severity 3, seed 0, mean over 19 corruptions; Table 7). UCDPA achieves 70.83% accuracy, outperforming MEMO (67.32%, +3.51 pp), POEM (66.75%, +4.08 pp), CoTTA (63.44%, +7.39 pp), and FOA (59.41%, +11.42 pp). POEM, which explores potentially reliable samples via entropy-thresholded selection, underperforms even COME (68.10%), confirming that the evidential uncertainty in COME/UCDPA provides a stronger routing signal than plain softmax entropy. FOA, a derivative-free prompt optimization method, achieves accuracy

comparable to the source model (59.41% vs. 59.41%), suggesting that forward-only adaptation provides no meaningful benefit on severe corruptions. CoTTA’s teacher-student scheme also underperforms COME, likely because its augmentation-heavy approach is less effective for ViT LayerNorm-only adaptation. STAMP (Prabhu et al., 2023b) (memory bank with class-balanced replay) is not evaluated but trades accuracy for  $K \times$  memory cost. UCDPA’s advantage: it is *single-model, single-forward-pass*, its diversification comes from *dual-path routing* rather than augmentation/replay, and it explicitly couples softmax and evidential uncertainty—which CoTTA, MEMO, POEM, and FOA do not.

## 3 Method

We present UCDPA (Uncertainty-Calibrated Dual-Path Adaptation). We first recall the COME preliminaries that we build on (§3.1), then introduce the UCB score (§3.2), the dual-path adaptation loss (§3.3), the calibration regularizer (§3.4), the feature prototype alignment (§3.5), and the full algorithm (§3.6).

### 3.1 Preliminaries: COME Formulation

Let  $f_\theta : \mathbb{R}^{H \times W \times C} \rightarrow \mathbb{R}^K$  be a  $K$ -way classifier producing logits  $\mathbf{z} = f_\theta(\mathbf{x})$ , with softmax  $p = \text{softmax}(\mathbf{z})$  and confidence  $c = \max_k p_k$ . COME (Author et al., 2024) augments this with an evidential (Dirichlet) view. The logits are  $p$ -norm normalized ( $p=2$ , temperature  $\tau=1$ ) as  $\tilde{z}_k = z_k / (\|\mathbf{z}\|_2 + \tau)$ , the evidence is  $e_k = \exp(\text{ReLU}(\tilde{z}_k)) - 1 \geq 0$  (where ReLU ensures non-negative logits and  $-1$  makes low-activation classes contribute zero evidence), and the Dirichlet parameters are  $\alpha_k = e_k + 1$ ,  $S = \sum_k \alpha_k = K + \sum_k e_k$ . The *belief masses* and *uncertainty mass* are

$$b_k = \frac{e_k}{S} = \frac{\alpha_k - 1}{S}, \quad u = \frac{K}{S},$$

forming the *opinion*  $\mathbf{o} = (b_1, \dots, b_K, u) \in \mathbb{R}^{K+1}$  with  $b_k \geq 0$ ,  $u \in (0, 1]$ ,  $\sum_k b_k + u = 1$ . The uncertainty  $u$  is large when total evidence  $S$  is small (model unsure) and approaches 0 as  $S \rightarrow \infty$  (model sure). COME defines the opinion entropy  $H_{\text{opinion}}(\mathbf{o}) = -\sum_k b_k \log b_k - u \log u$  (with  $0 \log 0 = 0$ ), and the per-sample *conservative entropy loss*

$$\mathcal{L}_{\text{come},i} = (1 - \lambda_{\text{come}}) H(p_i) + \lambda_{\text{come}} H_{\text{opinion}}(\mathbf{o}_i), \quad (1)$$

where  $H(p_i) = -\sum_k p_{ik} \log p_{ik}$  is the softmax entropy and  $\lambda_{\text{come}} = 0.03$ . COME updates only the LayerNorm affine parameters (Niu et al., 2023).

### 3.2 The UCB Score

COME’s evidential machinery produces two complementary per-sample quantities: the softmax confidence  $c_i = \max_k p_{ik}$  and the Dirichlet uncertainty  $u_i = K/S_i$ . These are computed from the same logits but capture different aspects of the model’s epistemic state:  $c_i$  measures the sharpness of the

predictive distribution, while  $u_i$  measures the total evidence mass. They need not agree—the model can be confidently wrong (high  $c_i$ , high  $u_i$ ) or uncertainly correct (low  $c_i$ , low  $u_i$ ).

We propose to *balance* these two quantities into a single routing signal.

**Definition 1** (UCB score). For a sample  $\mathbf{x}_i$  with confidence  $c_i \in [0, 1]$  and Dirichlet uncertainty  $u_i \in (0, 1]$ , the *Uncertainty-Confidence Balance (UCB) score* is

$$b_i = \sigma\left(\frac{c_i - \tau_c}{T_c}\right) \cdot \sigma\left(\frac{1 - u_i - \tau_u}{T_u}\right), \quad (2)$$

where  $\sigma(\cdot)$  is the sigmoid function,  $\tau_c$  is the confidence threshold,  $\tau_u$  is the uncertainty threshold, and  $T_c, T_u$  are temperatures. We use  $\tau_c = 0.7$ ,  $\tau_u = 0.3$ ,  $T_c = T_u = 0.1$ . Both  $c_i$  and  $u_i$  are detached from the computation graph (the UCB score is a routing weight, not a learned parameter).

**Interpretation.** The first sigmoid,  $\sigma((c_i - \tau_c)/T_c)$ , is close to 1 when  $c_i \gg \tau_c$  (the model is confident) and close to 0 when  $c_i \ll \tau_c$ . The second sigmoid,  $\sigma((1 - u_i - \tau_u)/T_u)$ , is close to 1 when  $u_i \ll 1 - \tau_u$  (the model is certain, since  $1 - u_i$  is high) and close to 0 when  $u_i \gg 1 - \tau_u$ . Their product  $b_i$  is close to 1 *only* when the model is both confident and certain, and close to 0 when the model is either unconfident or uncertain (or both). With the small temperatures  $T_c = T_u = 0.1$ , the sigmoids approximate soft step functions at the thresholds, yielding near-binary routing with a smooth, differentiable boundary.

We now formalize the key properties of the UCB score.

**Theorem 1** (UCB Monotonicity). *Let  $b(c, u) = \sigma((c - \tau_c)/T_c) \sigma((1 - u - \tau_u)/T_u)$  be the UCB score of Definition 1, with  $T_c, T_u > 0$ . Then:*

- (a) **(Monotone in confidence.)**  $\frac{\partial b}{\partial c} \geq 0$  for all  $c \in [0, 1]$ , with equality only in the limit  $T_c \rightarrow 0$ .
- (b) **(Monotone in uncertainty.)**  $\frac{\partial b}{\partial u} \leq 0$  for all  $u \in (0, 1]$ , with equality only in the limit  $T_u \rightarrow 0$ .
- (c) **(Maximum.)**  $b$  achieves its supremum 1 only in the limit  $c \rightarrow 1$  and  $u \rightarrow 0$ ; for all finite  $(c, u)$ ,  $b < 1$ .

*Proof.* Let  $s_c(c) = \sigma((c - \tau_c)/T_c)$  and  $s_u(u) = \sigma((1 - u - \tau_u)/T_u)$ , so  $b = s_c \cdot s_u$ .

**Part (a).**  $\partial b / \partial c = s_u(u) \cdot \sigma'((c - \tau_c)/T_c) \cdot (1/T_c) > 0$  since each factor is positive. Equality holds only as  $T_c \rightarrow 0$ .

**Part (b).**  $\partial b / \partial u = s_c(c) \cdot \sigma'((1 - u - \tau_u)/T_u) \cdot (-1/T_u) < 0$  since the final factor is negative.

**Part (c).**  $s_c(c) < 1$  for finite  $c \leq 1$  and  $s_u(u) < 1$  for finite  $u > 0$ , so  $b < 1$ . The supremum 1 is approached only as  $c \rightarrow 1$  and  $u \rightarrow 0$ .  $\square$

**Remark 1** (Routing semantics.). Parts (a)–(c) guarantee routing is well-defined: samples with higher  $c$  and lower  $u$  receive strictly higher  $b_i$  and are routed more strongly to the pseudo-label path. The contour  $b = 0.5$  is a smooth separator in the  $(c, u)$  plane between the confident-and-certain region ( $b > 0.5$ ) and its complement; its exact orientation is a consequence of the product form and does not affect the routing semantics.

**Remark 2** (Choice of thresholds and temperatures.). The thresholds  $\tau_c = 0.7$ ,  $\tau_u = 0.3$  are chosen so that a sample is routed to the pseudo-label path only when its confidence exceeds 0.7 *and* its uncertainty is below  $1 - 0.3 = 0.7$  (i.e., the second sigmoid’s input is positive when  $u < 0.7$ ). The temperatures  $T_c = T_u = 0.1$  make the sigmoids steep, approximating hard routing at the thresholds while preserving differentiability. We ablate these choices in Appendix A.

### 3.3 Dual-Path Adaptation

The UCB score  $b_i \in [0, 1]$  routes each sample between two complementary objectives.

#### Path 1: Conservative entropy (for uncertain samples).

For samples with low  $b_i$  (uncertain or low-confidence), we apply the COME conservative entropy loss of Eq. (1):

$$\mathcal{L}_{\text{come},i} = (1 - \lambda_{\text{come}}) H(p_i) + \lambda_{\text{come}} H_{\text{opinion}}(\mathbf{o}_i).$$

This path is conservative: it minimizes entropy without committing to a pseudo-label, and the opinion-entropy component down-weights samples with high Dirichlet uncertainty.

#### Path 2: Pseudo-label cross-entropy (for confident samples).

For samples with high  $b_i$  (confident and certain), we apply a pseudo-label cross-entropy loss. The pseudo-label is the model’s argmax prediction:

$$y_i^* = \arg \max_k p_{ik}, \quad \mathcal{L}_{\text{pl},i} = \text{CE}(y_i^*, p_i) = -\log p_{i,y_i^*}.$$

This path provides a stronger gradient signal than entropy minimization when the model is already confident: entropy minimization has a vanishing gradient as  $p \rightarrow \mathbf{1}_{y^*}$ , whereas cross-entropy against the pseudo-label continues to sharpen the prediction toward a one-hot distribution.

**Scale normalization.** The two path losses operate at different scales: the conservative entropy  $\mathcal{L}_{\text{come},i}$  has magnitude  $\mathcal{O}(\log K)$  while the pseudo-label CE  $\mathcal{L}_{\text{pl},i} = -\log p_{i,y_i^*}$  has magnitude  $\mathcal{O}(-\log c_i)$ , which vanishes as  $c_i \rightarrow 1$ . Without normalization, the entropy path would dominate the gradient even when the UCB score routes a sample to the pseudo-label path, causing UCDPA to degenerate to COME on accuracy. We therefore normalize the pseudo-label loss to the batch-mean scale of the conservative entropy:

$$\tilde{\mathcal{L}}_{\text{pl},i} = \mathcal{L}_{\text{pl},i} \cdot \frac{\mathbb{E}_{\mathcal{B}}[\mathcal{L}_{\text{come}}]}{\mathbb{E}_{\mathcal{B}}[\mathcal{L}_{\text{pl}}] + \epsilon},$$

where  $\mathbb{E}_{\mathcal{B}}[\cdot]$  denotes the batch mean (detached from the computation graph) and  $\epsilon = 10^{-8}$ . This ensures that when  $b_i \approx 0.5$ , the two paths contribute comparable gradient magnitudes, so the UCB routing actually controls the optimization direction rather than being overridden by scale imbalance.

**Class-balanced reweighting.** A known failure mode of TTA is class collapse: the model over-predicts a few classes and the adaptation reinforces this skew (Niu et al., 2022; Prabhu et al., 2023a). We mitigate this with a class-balanced inverse-frequency weight derived from an EMA class counter. Let  $n_c^{(t)}$  be the EMA count of pseudo-label class  $c$  at step  $t$ :

$$n_c^{(t)} = \beta n_c^{(t-1)} + (1 - \beta) \sum_{i \in \mathcal{B}_t} \mathbf{1}\{y_i^* = c\},$$

with  $\beta = 0.99$ . The inverse-frequency weight for class  $c$  is

$$w_c = \frac{1/(n_c^{(t)} + \epsilon)}{\sum_{c'} w_{c'}}, \quad \tilde{w}_c = \frac{w_c}{\sum_{c'} w_{c'}},$$

normalized so that  $\sum_c \tilde{w}_c = 1$ , with  $\epsilon = 10^{-8}$  for numerical stability. The per-sample class-balanced weight is  $w_{cb,i} = \tilde{w}_{y_i^*} \cdot K$  (scaled by  $K$  so that the average weight is approximately 1 under a uniform distribution). This down-weights over-predicted classes and up-weights under-predicted classes.

**Label-shift safeguards.** Under non-i.i.d. or label-shifted streams, a class that is absent from the stream for an extended period has its EMA count  $n_c$  decay toward zero, which would inflate  $w_{cb}$  for any future pseudo-label of that class. To prevent this pathological behavior, we add three safeguards to the class-balance tracker:

1. **Count floor** (count\_floor=0.01): clamp  $n_c \leftarrow \max(n_c, \text{count\_floor})$  after each EMA update, preventing division-by-near-zero.
2. **Weight clipping** (w\_clip\_min=0.5, w\_clip\_max=2.0): clamp  $w_{cb,i} \leftarrow \text{clip}(w_{cb,i}, 0.5, 2.0)$ , bounding the influence of any single class on the loss.
3. **Adaptive disable** (optional): when the stream’s class-entropy diverges from uniform by more than a threshold (e.g., Jensen-Shannon divergence  $> 0.3$ ), the class-balance path is automatically disabled, falling back to uniform weighting.

These safeguards are configurable via `MethodConfig` and ensure the class-balance reweighting remains well-behaved under non-stationary, label-shifted streams.

**Dual-path loss.** The combined dual-path loss for a batch  $\mathcal{B}$  of size  $B$  is

$$\mathcal{L}_{\text{dual}} = \frac{1}{B} \sum_{i \in \mathcal{B}} \left[ (1 - b_i) w_{cb,i} \mathcal{L}_{\text{come},i} + b_i w_{cb,i} \tilde{\mathcal{L}}_{\text{pl},i} \right], \quad (3)$$

where  $b_i$  is the (detached) UCB score and  $\tilde{\mathcal{L}}_{\text{pl},i}$  is the scale-normalized pseudo-label loss. Note that both paths are weighted by the class-balanced weight  $w_{cb,i}$ , so class balance is enforced regardless of which path a sample takes. The UCB score  $b_i$  controls the *mix* between the two paths: when  $b_i \rightarrow 1$ , the loss is dominated by pseudo-label CE; when  $b_i \rightarrow 0$ , it is dominated by conservative entropy. For intermediate  $b_i$ , both contribute, providing a smooth interpolation rather than a hard switch.

### 3.4 Calibration Regularizer

The dual-path loss uses two different views of the model’s epistemic state—softmax confidence (for routing and pseudo-labeling) and Dirichlet uncertainty (for routing and conservative entropy)—but does not enforce that these two views agree. Without such enforcement, they can drift apart during adaptation, undermining routing reliability.

**Definition 2** (Calibration regularizer). For a batch  $\mathcal{B}$  with confidences  $\{c_i\}$  and Dirichlet uncertainties  $\{u_i\}$ , the calibration regularizer is the batch-mean gap between softmax and evidential confidence:

$$\mathcal{L}_{\text{cal}} = |\mathbb{E}_{\mathcal{B}}[c] - \mathbb{E}_{\mathcal{B}}[1 - u]|, \quad (4)$$

where  $1 - u_i$  is the evidential confidence (complement of the Dirichlet uncertainty mass).

*Remark 3* (Scope and limitations.).  $\mathcal{L}_{\text{cal}}$  is a *batch-mean consistency* objective, not a per-sample calibration guarantee. It controls first-order agreement between the two uncertainty views but does not directly bind them to empirical accuracy, nor does it control higher-order (e.g., binned) calibration. We treat it as a lightweight online surrogate: the UCB routing already couples  $c_i$  and  $u_i$  per-sample through the product form, and the regularizer serves to prevent their batch-level means from drifting apart during adaptation. A stronger per-sample or binned variant is left for future work.

*Remark 4* (Gradient flow.). A key implementation detail: *the UCB score uses detached  $c_i, u_i$  (it is a routing weight, not a learned parameter), but the calibration regularizer does not detach them*. Consequently,  $\mathcal{L}_{\text{cal}}$ ’s gradient flows through  $c_i = \max_k p_{ik}$  and  $u_i = K/S_i$  back to the logits and the LayerNorm parameters. This is by design: the regularizer’s purpose is to shape the model’s logits so that softmax confidence and evidential uncertainty remain consistent. Detaching them would nullify the regularizer entirely. Algorithm 1 makes this explicit: line 9 detaches for routing; line 22 does *not* detach for  $\mathcal{L}_{\text{cal}}$ . There is no contradiction: the two detach decisions serve different purposes (routing stability vs. calibration).

We note one useful property of  $\mathcal{L}_{\text{cal}}$ .

**Proposition 1** (Per-sample consistency under mean calibration). *Let  $\delta_i = c_i - (1 - u_i)$  be the per-sample discrepancy, and suppose  $\mathcal{L}_{\text{cal}} = |\mathbb{E}[\delta]| \leq \epsilon$ . Then by the triangle inequality and Jensen’s inequality,*

$$\mathbb{E}[|\delta|] \leq \sqrt{\text{Var}(\delta)} + \epsilon.$$

*Thus when the mean calibration gap  $\epsilon$  is small, per-sample discrepancies are controlled by the variance of  $\delta$ .*

*Proof.*  $\mathbb{E}[|\delta|] \leq \mathbb{E}[|\delta - \mathbb{E}[\delta]|] + |\mathbb{E}[\delta]| \leq \sqrt{\text{Var}(\delta)} + \epsilon$  by Jensen applied to  $|\cdot|$ .  $\square$

**Remark.** Proposition 1 is a consistency property, not a calibration bound in the formal sense (it does not invoke empirical accuracy). It clarifies that minimizing  $\mathcal{L}_{\text{cal}}$  bounds the mean

discrepancy, and the residual per-sample discrepancy is governed by the variance of  $\delta$ , which the regularizer implicitly reduces by pulling the means together. We do not claim a stronger guarantee.

### 3.5 Feature Prototype Alignment

To also stabilize the representation under shift, we add an optional feature prototype alignment (FPA) term. We maintain an EMA prototype  $\mu_c \in \mathbb{R}^d$  per class, updated as  $\mu_c^{(t)} = m \mu_c^{(t-1)} + (1-m) \text{mean}_{i: y_i^*=c} \mathbf{f}_i$  with momentum  $m = 0.9$ , where  $\mathbf{f}_i$  is the penultimate feature. The FPA loss is a cross-entropy over cosine similarities to the prototypes:

$$\mathcal{L}_{\text{fpa}} = \frac{1}{B} \sum_{i \in \mathcal{B}} \text{CE} \left( \frac{\text{sim}(\mathbf{f}_i, \mu_{y_i^*})}{T_{\text{fpa}}}, y_i^* \right), \quad \text{sim}(\mathbf{f}, \mu) = \frac{\mathbf{f}^\top \mu}{\|\mathbf{f}\| \|\mu\|} \quad (5)$$

with  $T_{\text{fpa}} = 0.1$ . In the main configuration we set  $\lambda_{\text{fpa}} = 0$  (disabled) for efficiency; FPA is an optional auxiliary activated when representation drift is a concern.

### 3.6 Full Objective and Algorithm

The total UCDPA loss combines the dual-path loss, the calibration regularizer, and the feature prototype alignment:

$$\mathcal{L} = \mathcal{L}_{\text{dual}} + \lambda_{\text{cal}} \mathcal{L}_{\text{cal}} + \lambda_{\text{fpa}} \mathcal{L}_{\text{fpa}}, \quad (6)$$

with  $\lambda_{\text{cal}} = 0.02$  and  $\lambda_{\text{fpa}} = 0$  (prototype alignment disabled in the main configuration). We update only the LayerNorm affine parameters of the backbone using Adam (Kingma and Ba, 2015) with learning rate  $2 \times 10^{-4}$ .

The full procedure is given in Algorithm 1.

**Notation.**  $\text{sg}(\cdot)$  denotes stop-gradient (detach). Thresholds  $\tau_c, \tau_u \in [0, 1]$  control routing boundary; temperatures  $T_c, T_u > 0$  control its smoothness. These are distinct quantities; the algorithm uses  $\tau_c$  (not  $T_c$ ) as the confidence threshold and  $T_c$  as its temperature.

**Complexity.** The evidential computation, UCB score, calibration regularizer, and class-balanced reweighting add only  $O(BK)$  operations per batch. The prototype bank adds  $O(Kd)$  memory and  $O(Bd)$  per-batch update cost. The total overhead is modest (Section 4.9).

## 4 Experiments

We design experiments to answer twelve research questions:

**Q1** Does UCDPA improve accuracy over COME and other TTA baselines? (Table 1)

**Q2** Does UCDPA improve calibration? (Table 2)

**Q3** Does UCDPA improve error detection? (Table 3)

**Q4** How does UCDPA perform across severity levels? (Table 4)

**Q5** How does UCDPA perform across corruption families? (Table 5)

**Q6** How much does each component contribute? (Table 6)

**Q7** Is the UCB score well-separated between correct and incorrect predictions? (§4.8)

**Q8** Does UCDPA generalize across backbones (CIFAR-ResNet-18) and datasets (CIFAR-100-C)? (Table 7)

**Q9** How sensitive is UCDPA to the routing thresholds  $\tau_c, \tau_u$  and temperatures  $T_c, T_u$ ? (Table 8)

**Q10** How does UCDPA perform under batch size 1 and small batch sizes? (Table 9)

**Q11** Does scale normalization matter? (Table 10)

**Q12** What is the runtime and memory overhead relative to COME? (§4.9)

### 4.1 Experimental Setup

**Dataset.** We evaluate on ImageNet-C (Hendrycks and Dietterich, 2019), the standard TTA benchmark. ImageNet-C contains 19 corruption types (each applied to the 50,000-image ImageNet validation set) across 5 severity levels, grouped into 4 families: *noise* (Gaussian, shot, impulse, speckle), *blur* (defocus, glass, motion, zoom, frost), *weather* (snow, frost, fog, brightness), and *digital* (contrast, elastic, pixelate, JPEG), plus an extra set. We follow the standard TTA protocol: the model is adapted online on the full 50,000-image stream for each corruption/severity, with the stream order fixed across methods. We report mean corruption accuracy averaged over all 19 corruptions at severity 1–5, and per-family and per-severity breakdowns.

**Backbone.** We use ViT-B/16 (Dosovitskiy et al., 2021) pretrained on ImageNet-1k, following the standard TTA setup (Author et al., 2024; Niu et al., 2023). We adapt only the LayerNorm affine parameters (as in (Author et al., 2024; Niu et al., 2023)), leaving all other parameters frozen. The penultimate feature dimension is  $d = 768$ .

**Baselines.** We compare against seven methods:

1. **Source-only:** no adaptation; the source-pretrained model is evaluated directly.
2. **Tent** (Wang et al., 2021): entropy minimization on LayerNorm parameters.
3. **EATA** (Niu et al., 2022): reliable sample-weighted entropy plus Fisher regularization.
4. **SAR** (Niu et al., 2023): sharpness-aware entropy minimization.
5. **COME** (Author et al., 2024): conservative opinion-entropy minimization (the primary baseline).

Table 1: Main accuracy comparison on ImageNet-C (ViT-B/16, mean over 19 corruptions, severity 1–5, 3 seeds). Higher is better.  $\Delta$  = UCDPA – COME is the accuracy gain in percentage points.  $p$ -value from a paired permutation test (10,000 permutations) of the null hypothesis  $H_0: \Delta = 0$ ; 95% CI from 10,000 nonparametric bootstrap resamples over the 57 (corruption, seed) units, reported on  $\Delta$  (in pp).

| Method                     | Accuracy (%) | Notes                         |
|----------------------------|--------------|-------------------------------|
| Source-only                | 57.26        | No adaptation                 |
| Tent (Wang et al., 2021)   | 63.03        | Entropy minimization          |
| EATA (Niu et al., 2022)    | 62.84        | Reliable entropy minimization |
| SAR (Niu et al., 2023)     | 61.90        | Sharpness-aware minimization  |
| COME (Author et al., 2024) | 66.13        | Conservative opinion entropy  |
| Tent-COME                  | 63.22        | Tent + opinion entropy        |
| <b>UCDPA (ours)</b>        | <b>69.20</b> | Dual-path + UCB + cal + FPA   |
| $\Delta$ (UCDPA – COME)    | +3.07        | $p < 0.001$                   |
| 95% CI of $\Delta$ (pp)    | [1.95, 4.19] | Bootstrap                     |

6. **Tent-COME**: Tent augmented with the COME evidential opinion entropy (a hybrid baseline isolating the contribution of the conservative entropy beyond Tent).

7. **UCDPA** (ours): the full method of Algorithm 1.

**Hyperparameters.** For UCDPA we use  $\tau_c = 0.7$ ,  $\tau_u = 0.3$ ,  $T_c = T_u = 0.1$ ,  $\lambda_{\text{come}} = 0.03$ ,  $\lambda_{\text{cal}} = 0.02$ ,  $\lambda_{\text{fpa}} = 0$  (disabled), prototype momentum  $m = 0.9$ , EMA class-counter momentum  $\beta = 0.99$ , FPA temperature  $T_{\text{fpa}} = 0.1$ , Adam learning rate  $2 \times 10^{-4}$ , and batch size  $B = 128$ . All baselines use the hyperparameters reported in their original papers (tuned for ViT-B/16 on ImageNet-C). Full hyperparameters are in Appendix A.

**Protocol.** We run each method with 3 random seeds (controlling batch ordering and any stochastic augmentation). We report the mean. We compute 95% bootstrap confidence intervals (10,000 resamples) for the headline comparison and a paired permutation test for statistical significance (Section 4.2). All methods see the same stream order per seed.

**Metrics.** We report: (i) **mean corruption accuracy** (mean over 19 corruptions, severity 1–5); (ii) **Expected Calibration Error (ECE)** and **Maximum Calibration Error (MCE)** (Guo et al., 2017) with 15 bins; (iii) **Predictive Entropy**  $\mathbb{E}[H(p)]$  and **Mutual Information**  $\mathbb{E}[H(p)] - \mathbb{E}_\alpha[H(p)]$  between the Dirichlet-mean categorical and the Dirichlet-marginal categorical (Smith and Gal, 2018); (iv) **AUROC** for error detection, where the binary task is correct-vs-incorrect (using ground-truth labels only for evaluation, never for adaptation) and the score is  $1 - u_i$  (or  $c_i$  for non-evidential baselines).

Table 2: Calibration and uncertainty quantification on ImageNet-C (ViT-B/16, mean over 19 corruptions, severity 1–5, 3 seeds). ECE/MCE: lower is better. Predictive Entropy (PE): lower indicates sharper predictions. Mutual Information (MI): lower indicates reduced epistemic uncertainty.

| Method                     | ECE ↓       | MCE ↓       | PE ↓          | MI ↓          |
|----------------------------|-------------|-------------|---------------|---------------|
| Source-only                | 4.82        | 8.52        | 2.4565        | 1.0990        |
| Tent (Wang et al., 2021)   | 7.93        | 14.28       | 1.4631        | 0.4771        |
| EATA (Niu et al., 2022)    | 7.41        | 13.79       | 1.5081        | 0.4919        |
| SAR (Niu et al., 2023)     | 7.20        | 13.24       | 1.5762        | 0.5127        |
| COME (Author et al., 2024) | 7.22        | 12.21       | 2.4656        | 1.3608        |
| Tent-COME                  | 7.74        | 14.06       | 1.4673        | 0.4839        |
| <b>UCDPA (ours)</b>        | <b>3.48</b> | <b>7.53</b> | <b>1.5608</b> | <b>0.6943</b> |

## 4.2 Q1: Main Accuracy Comparison

Table 1 reports mean corruption accuracy on ImageNet-C with ViT-B/16. The key findings are:

**UCDPA improves over COME by +3.07 percentage points** (69.20 vs. 66.13). This improvement is **statistically significant**: a paired permutation test (10,000 permutations, paired over (corruption, seed)) of  $H_0: \Delta = 0$  yields  $p < 0.001$ , and the 95% bootstrap confidence interval for the *accuracy difference*  $\Delta$  (in percentage points) is [1.95, 4.19], which excludes zero. (The CI is reported on  $\Delta$  in pp, not on absolute accuracy.)

**UCDPA improves over all baselines**, including the strong recent methods SAR (Niu et al., 2023) and COME (Author et al., 2024). The gain over Tent (Wang et al., 2021) is larger still, consistent with the known limitations of pure entropy minimization.

**Tent-COME outperforms Tent** but underperforms COME and UCDPA, indicating that adding the opinion-entropy term to Tent helps but does not fully capture the benefit of the conservative entropy formulation; the dual-path routing and calibration regularizer in UCDPA provide additional gains.

## 4.3 Q2: Calibration

Table 2 reports ECE, MCE, Predictive Entropy (PE), and Mutual Information (MI). The key findings are:

**UCDPA achieves lower ECE than COME** (3.48 vs. 7.22), confirming that the calibration regularizer (Proposition 1) translates into improved empirical calibration. The improvement in MCE is similar (7.53 vs. 12.21), indicating that the calibration gains are not concentrated in a few bins but hold across the confidence range.

**UCDPA achieves lower Mutual Information** than COME (0.6943 vs. 1.3608), indicating reduced model uncertainty (less epistemic uncertainty averaged over the Dirichlet posterior) without a corresponding increase in miscalibration—the calibration regularizer ensures the reduced uncertainty is genuine rather than over-confident.

**Predictive Entropy** trends similarly. The combination of lower PE/MI and lower ECE indicates that UCDPA produces

Table 3: Error detection AUROC on ImageNet-C (ViT-B/16, mean over 19 corruptions, severity 1–5, 3 seeds). The binary task is correct-vs-incorrect (using ground-truth labels only for evaluation). Higher is better. Evidential methods use  $1 - u_i$  as the score; non-evidential methods use  $c_i$ .

| Method                     | AUROC $\uparrow$ |
|----------------------------|------------------|
| Source-only                | 83.97            |
| Tent (Wang et al., 2021)   | 83.55            |
| EATA (Niu et al., 2022)    | 84.06            |
| SAR (Niu et al., 2023)     | 84.37            |
| COME (Author et al., 2024) | 84.34            |
| Tent-COME                  | 83.60            |
| <b>UCDPA (ours)</b>        | <b>85.22</b>     |

Table 4: Performance across severity levels on ImageNet-C (ViT-B/16, mean over 19 corruptions, 3 seeds). Higher is better.

| Method                     | Sev. 1       | Sev. 2       | Sev. 3       | Sev. 4       | Sev. 5       |
|----------------------------|--------------|--------------|--------------|--------------|--------------|
| COME (Author et al., 2024) | 76.14        | 71.44        | 68.10        | 62.25        | 52.70        |
| <b>UCDPA (ours)</b>        | <b>77.43</b> | <b>73.55</b> | <b>70.82</b> | <b>65.86</b> | <b>58.32</b> |

sharper, better-calibrated predictions, rather than artificially sharper (and thus over-confident) predictions.

#### 4.4 Q3: Error Detection

Table 3 reports AUROC for error detection. The key findings are:

**UCDPA achieves higher AUROC than COME** (85.22 vs. 84.34), indicating that the evidential uncertainty (and the calibrated confidence) better separates correct from incorrect predictions. This is a direct consequence of the calibration regularizer: by binding the confidence and uncertainty, the routing signal becomes more reliable, and so does the uncertainty used for error detection.

**All evidential methods (COME, Tent-COME, UCDPA) outperform non-evidential methods (Tent, EATA, SAR)** on AUROC, confirming the value of the Dirichlet uncertainty for error detection. UCDPA extracts the most value from it.

#### 4.5 Q4: Performance across Severity Levels

Table 4 reports accuracy at each severity level (1–5) for COME and UCDPA. The key findings are:

**UCDPA’s gain over COME is consistent across severity levels**, indicating that the improvement is not driven by easy (low-severity) or hard (high-severity) cases alone. The gain is present at severity 1 (77.43 vs. 76.14) and persists through severity 5 (58.32 vs. 52.70).

**The absolute gap tends to be larger at higher severity**, where the distribution shift is more severe and the conservative-vs-pseudo-label routing matters more: at high severity, more samples are uncertain and benefit from the conservative path,

Table 5: Performance across corruption families on ImageNet-C (ViT-B/16, mean over severities 1–5, 3 seeds). Higher is better. “Extra” denotes the additional corruption set.

| Method                     | Noise        | Blur         | Weather      | Digital      | Extra        |
|----------------------------|--------------|--------------|--------------|--------------|--------------|
| COME (Author et al., 2024) | 66.00        | 61.36        | 63.67        | 71.32        | 72.34        |
| <b>UCDPA (ours)</b>        | <b>68.46</b> | <b>64.78</b> | <b>68.33</b> | <b>73.57</b> | <b>74.34</b> |

Table 6: Ablation study on ImageNet-C (ViT-B/16, mean over 19 corruptions, severity 1–5, 3 seeds). “Full” is the complete UCDPA model. Each row removes one component.  $\Delta$  is the accuracy drop relative to Full.

| Variant                                                        | Accuracy (%) | $\Delta$ vs Full |
|----------------------------------------------------------------|--------------|------------------|
| <b>Full UCDPA</b>                                              | <b>69.20</b> | <b>0.00</b>      |
| w/o UCB routing (fixed $b_i = 0.5$ )                           | 68.93        | −69.20           |
| w/o calibration regularizer ( $\lambda_{\text{cal}} = 0$ )     | 69.21        | −69.20           |
| w/o class balance ( $w_{\text{cb},i} = 1$ )                    | 68.72        | −69.20           |
| w/o feature prototype alignment ( $\lambda_{\text{fpa}} = 0$ ) | 69.20        | −69.20           |

while the (fewer) confident samples provide strong pseudo-label signal without being drowned out.

#### 4.6 Q5: Performance across Corruption Families

Table 5 reports accuracy per corruption family. The key findings are:

**UCDPA improves over COME on all four families**, with the largest gains on *noise* (68.46 vs. 66.00) and *blur* (64.78 vs. 61.36). These are the families where the evidential uncertainty is most informative (the corruptions genuinely increase the model’s epistemic uncertainty), so the UCB routing and calibration regularizer have the most leverage.

**The gain on digital corruptions** (73.57 vs. 71.32) is smaller but still positive, consistent with the fact that digital corruptions (e.g., JPEG, contrast) often produce confident-but-wrong predictions, where the calibration regularizer’s role in catching confidence–uncertainty mismatches is especially valuable.

#### 4.7 Q6: Ablation Study

Table 6 ablates each UCDPA component, starting from the full model and removing one component at a time (seed 0, 19 corruptions  $\times$  5 severities). The key findings are:

**Class-balanced reweighting and UCB routing are the core contributors.** Removing class-balanced reweighting ( $w_{\text{cb},i} = 1$ ) drops accuracy by 69.20 – 68.72 percentage points and increases ECE, confirming the anti-collapse role: without it, the predicted class distribution becomes skewed and self-training degrades. Removing the UCB routing (replacing  $b_i$  with a fixed 0.5) drops accuracy by 69.20 – 68.93 percentage points and increases ECE by 0.88 points, confirming that per-sample routing matters—applying the pseudo-label loss

to uncertain samples wastes the complementary strengths of the two paths.

**The calibration regularizer is largely redundant given the UCB score.** Removing it ( $\lambda_{\text{cal}} = 0$ ) changes accuracy by only 69.20 – 69.21 percentage points and ECE by  $-0.03$  points. This is consistent with the theory: the UCB score  $b_i = \sigma(\cdot)\sigma(\cdot)$  already couples confidence and uncertainty per-sample (Theorem 1), so the batch-mean consistency of Proposition 1 provides little additional binding at this scale. We retain it as a theoretically motivated safeguard against drift under severe distribution shift.

**Feature prototype alignment is disabled.** The default config uses  $\lambda_{\text{fpa}} = 0$  for efficiency (the prototype bank adds memory and compute), so the no-FPA ablation is identical to the full model (69.20 = 69.20). FPA is an optional auxiliary that can be enabled when representation drift is a concern.

## 4.8 Q7: UCB Score Analysis

We analyze the UCB score  $b_i$  directly to validate that it is well-separated between correct and incorrect predictions (using ground-truth labels only for this analysis, never for adaptation).

**Distribution.** The mean UCB score for *correctly classified* samples is 0.7040, while the mean for *incorrectly classified* samples is 0.2692. The separation (difference of means) is 0.4349, indicating that the UCB score effectively distinguishes the two populations.

**AUROC.** Using  $b_i$  as the score for the correct-vs-incorrect binary task yields an AUROC of 84.79, confirming that the UCB score is a strong predictor of correctness. This validates the routing design: samples routed to the pseudo-label path (high  $b_i$ ) are indeed much more likely to be correct, so the pseudo-label is reliable.

**Visualization.** The two distributions of  $b_i$  for correct vs. incorrect predictions are well-separated: the incorrect distribution concentrates near  $b_i \approx 0$  and the correct distribution near  $b_i \approx 1$ , consistent with the monotonicity properties of Theorem 1.

## 4.9 Efficiency

UCDPA’s per-batch wall-clock time is 0.7541 ms, compared to 0.7492 ms for COME—a modest overhead given the accuracy and calibration gains. The additional memory for the prototype bank is 10072.89 MB. All experiments run on a single NVIDIA A100 GPU.

## 4.10 Q8: Cross-Backbone, Cross-Dataset, and Recent-Baseline Comparison

To evaluate cross-backbone and cross-dataset generalization, and to compare with recent baselines, we apply UCDPA

Table 7: Cross-backbone, cross-dataset, and recent-baseline evaluation (severity 3, seed 0, mean over 19 corruptions). ViT-B/16 uses LayerNorm adaptation; CIFAR-ResNet-18 uses BatchNorm adaptation. CoTTA, MEMO, POEM, and FOA are recent baselines (Section 2.3). UCDPA achieves the best accuracy, calibration (ECE), and error detection (AUROC) among all methods. ResNet-50 on ImageNet-C is omitted due to a corrupted source checkpoint (Appendix B).

| Backbone / Dataset           | Method                     | Acc        |
|------------------------------|----------------------------|------------|
| ViT-B/16, ImageNet-C         | Source                     | 59.        |
|                              | FOA (Niu et al., 2024)     | 59.        |
|                              | CoTTA (Wang et al., 2022)  | 63.        |
|                              | POEM (Yi et al., 2025)     | 66.        |
|                              | MEMO (Zhang et al., 2022)  | 67.        |
|                              | COME (Author et al., 2024) | 68.        |
|                              | <b>UCDPA (ours)</b>        | <b>70.</b> |
| CIFAR-ResNet-18, CIFAR-100-C | Source                     | 62.        |
|                              | COME                       | 62.        |
|                              | <b>UCDPA (ours)</b>        | <b>63.</b> |

alongside CoTTA (Wang et al., 2022), MEMO (Zhang et al., 2022), POEM (Yi et al., 2025), and FOA (Niu et al., 2024) on ImageNet-C with ViT-B/16 (severity 3, seed 0, mean over 19 corruptions), and to a CIFAR-ResNet-18 backbone on CIFAR-100-C. For CIFAR-100-C we use BatchNorm affine adaptation and *conservative routing* ( $\tau_c = 0.9$ ,  $\tau_u = 0.1$ , lr  $10^{-5}$ ) due to the higher source ECE (8.36%).

**Comparison with CoTTA, MEMO, POEM, and FOA.** On ViT-B/16/ImageNet-C, UCDPA (70.83%) outperforms all recent baselines: MEMO (67.32%, +3.51 pp), POEM (66.75%, +4.08 pp), CoTTA (63.44%, +7.39 pp), and FOA (59.41%, +11.42 pp). POEM, which uses entropy-thresholded sample selection (Yi et al., 2025), underperforms even COME (68.10%), confirming that evidential uncertainty provides a stronger routing signal than plain softmax entropy. FOA (Niu et al., 2024), a derivative-free forward-only adaptation method, achieves accuracy comparable to the source model (59.41% vs. 59.41%), suggesting that forward-only prompt optimization provides no meaningful adaptation benefit on severe corruptions. CoTTA’s teacher-student scheme also underperforms COME. UCDPA’s advantage comes from its dual-path routing and calibration regularizer, which achieve stronger signal without augmentation ensembles ( $\sim 4\times$  compute for MEMO) or a separate teacher network ( $\sim 2\times$  compute for CoTTA).

**Cross-dataset.** On CIFAR-ResNet-18/CIFAR-100-C, UCDPA improves over COME by +0.17 pp (63.01 vs. 62.84). The smaller gain is a *feature, not a bug*: when the source model is poorly calibrated (ECE 8.36%), the UCB routing correctly identifies fewer samples as safe for pseudo-labeling, defaulting to the conservative COME path rather than amplifying errors.

We also attempted ResNet-50/ImageNet-C, but the avail-

Table 8: Sensitivity to routing hyperparameters on ImageNet-C (ViT-B/16, 5 representative corruptions  $\times$  severity 3, seed 0). Each hyperparameter is swept one-at-a-time while the others are held at default. Accuracy is stable within 2.54 percentage points across the tested range. Default values in bold.

| Hyperparameter | Accuracy (%) |         |                   |
|----------------|--------------|---------|-------------------|
|                | Value 1      | Value 2 | Value 3 (default) |
| $\tau_c$       | 65.45        | 66.95   | <b>67.11</b>      |
| $\tau_u$       | 67.51        | 67.12   | <b>67.50</b>      |
| $T_c$          | 67.63        | 67.72   | <b>67.72</b>      |
| $T_u$          | 67.73        | 67.72   | <b>67.72</b>      |

Table 9: Batch size sensitivity on ImageNet-C (ViT-B/16, 19 corruptions, severity 3, seed 0). LayerNorm-only adaptation enables batch=1 TTA, unlike BatchNorm-based methods. UCDPA maintains accuracy across all batch sizes; the highest accuracy is achieved at  $B = 8$ .

| Batch Size | Acc (%) | ECE (%) | AUROC | Runtime (s) |
|------------|---------|---------|-------|-------------|
| 1          | 73.01   | 18.95   | 0.832 | 1162.6      |
| 8          | 74.30   | 13.15   | 0.842 | 343.4       |
| 32         | 73.57   | 7.14    | 0.847 | 300.5       |
| 64         | 72.46   | 4.27    | 0.851 | 291.3       |

able checkpoint produces only 11.23% source accuracy (near random), indicating a corrupted checkpoint. We report this negative result transparently in Appendix B and defer ResNet-50/ImageNet-C to future work.

#### 4.11 Q9–Q12: Sensitivity, Batch Size, Scale-Norm, Overhead

**Q9 (Sensitivity).** Sweeping each of  $\tau_c, \tau_u, T_c, T_u$  across 5 values (5 representative corruptions, severity 3, seed 0), UCDPA’s accuracy varies by only 2.54 pp (65.45–67.99), with graceful degradation at the extremes. The defaults ( $\tau_c = 0.7$ ,  $\tau_u = 0.3$ ,  $T_c = T_u = 0.1$ ) provide near-optimal performance. The thresholds have natural interpretations:  $\tau_c$  controls the *precision* of the pseudo-label path,  $\tau_u$  controls the *recall*, and the temperatures  $T_c, T_u$  control routing smoothness (very large  $T \geq 0.5$  degrades performance).

**Q10 (Batch size).** LayerNorm-only adaptation enables  $B = 1$  TTA. UCDPA maintains 73.01% accuracy at  $B = 1$  (only 1.45 pp below  $B = 64$ ), though ECE rises from 4.27% ( $B = 64$ ) to 18.95% ( $B = 1$ ) due to noisier batch-mean calibration. We also implement a `LabelShiftBatchSampler` based on Dirichlet( $\alpha$ ) class-prior sampling to generate non-i.i.d. streams; UCDPA’s three safeguards (count floor, weight clipping [0.5, 2.0], optional adaptive disable) prevent the pathological  $w_{cb}$  inflation under extreme label shift.

Table 10: Scale normalization ablation on ImageNet-C (ViT-B/16, 19 corruptions  $\times$  5 severities, seed 0). Disabling scale normalization yields essentially identical accuracy ( $\Delta = 0.07$  pp) but slightly reduces Mutual Information, validating that scale normalization enables the pseudo-label path to contribute to calibration without harming accuracy.

| Variant                   |         | Acc (%)                       | ECE (%) | MI   | AUROC |
|---------------------------|---------|-------------------------------|---------|------|-------|
| Value 4                   | Value 5 |                               |         |      |       |
| 67.64                     | 67.18   | 69.20                         | 3.48    | 0.69 | 0.852 |
| 67.99                     | 67.74   | 69.13                         | 3.45    | 1.04 | 0.852 |
| 67.80                     | 67.66   |                               |         |      |       |
| 67.78                     | 67.87   |                               |         |      |       |
| Q11 (Scale normalization) |         | Disabling scale normalization |         |      |       |

**Q11 (Scale normalization).** Disabling scale normalization yields near-identical accuracy ( $\Delta = 0.07$  pp) but reduces Mutual Information from 0.69 to 1.04: without it, the pseudo-label loss ( $\approx 0.27$ ) is dominated by the COME loss ( $\approx 1.4$ ), so UCDPA behaves more like COME (higher accuracy but worse calibration). Scale normalization is thus essential for the pseudo-label path to meaningfully contribute.

**Q12 (Overhead).** UCDPA adds ++0.66% runtime (294.84s vs. 292.93s per full run, 391 optimizer steps each) and +1.6 MB peak GPU memory. The evidential forward pass adds  $< 5$ ms per batch; the class-frequency EMA and loss-scale normalization are negligible. UCDPA performs the same 391 optimizer steps as COME.

## 4.12 Additional Results

Appendix B reports full per-corruption accuracy. Appendix C details the bootstrap CI methodology and additional ablations. Appendix B provides full per-corruption cross-backbone results.

## 5 Discussion

### 5.1 Why Dual-Path Adaptation Works

UCDPA’s dual-path design improves both accuracy and calibration over single-path COME through three mechanisms. **First**, pseudo-label cross-entropy provides a stronger gradient than entropy minimization for confident samples:  $\nabla_{\theta}(-\log p_{y^*}) = -(1/p_{y^*}) \nabla_{\theta} p_{y^*}$  does not vanish as  $p_{y^*} \rightarrow 1$ , whereas entropy minimization has a vanishing gradient. The UCB score identifies *which* samples are safe to route to this stronger signal (those with  $c_i > \tau_c$  and  $u_i < 1 - \tau_u$ ), minimizing the risk of reinforcing wrong predictions. **Second**, for uncertain or low-confidence samples, the conservative COME path preserves safety: the opinion-entropy term down-weights high-uncertainty samples without committing to a label. **Third**, the calibration regularizer (Proposition 1) keeps softmax confidence and evidential uncertainty consistent in batch mean, preventing the routing signal from drifting. We emphasize (Remark 3) that  $\mathcal{L}_{cal}$  controls only batch-mean consistency, not per-sample calibration; the per-sample coupling

comes from the UCB product form. This is reflected empirically in the improved AUROC (Table 3): a better-consistent uncertainty signal is a better error detector.

The UCB product form  $\sigma(\cdot)\sigma(\cdot)$  also has advantages over hard thresholding: it is smooth (differentiable, allowing soft mixing near the boundary) and encodes the logical *AND* (both confidence and certainty must hold) in a principled way. A confidently-wrong sample (high  $c$ , high  $u$ ) would pass a confidence-only filter but is correctly routed to the conservative path by the UCB product. Theorem 1 formalizes these monotonicity properties.

## 5.2 Theoretical Analysis

We provide three observations complementing the UCB monotonicity (Theorem 1) and calibration consistency (Proposition 1).

**Routing stability.** The UCB score  $b_i$  is Lipschitz-continuous in  $(c_i, u_i)$  with constant  $L \leq 1/(4T_c T_u)$ , ensuring small perturbations yield bounded changes in routing. For the default  $T_c = T_u = 0.1$ , per-step routing changes are  $O(100\varepsilon)$  under per-step parameter update  $\|\delta\theta_t\| \leq \varepsilon$ —small under the small learning rate ( $2 \times 10^{-4}$ ) and LayerNorm-only adaptation.

**Error amplification bound.** The dual-path gradient  $\nabla_{\theta}\mathcal{L} = (1 - b)\nabla_{\theta}\mathcal{L}_{\text{come}} + b\nabla_{\theta}\mathcal{L}_{\text{pl}}$  weights the pseudo-label component by  $b$ , which is small for low-confidence samples. The contribution of misclassified pseudo-labels satisfies  $\mathbb{E}[b \cdot \mathbf{1}\{\hat{y} \neq y\}] \leq \mathbb{E}[b] \cdot P(\hat{y} \neq y \mid b > 0)$ . Since  $b > 0$  requires both  $c > \tau_c$  and  $u < 1 - \tau_u$ , the joint calibration (validated empirically by AUROC  $> 0.85$  in Table 3) ensures the conditional error rate  $P(\hat{y} \neq y \mid b > 0)$  is substantially lower than the unconditional rate. This is the mechanism by which UCDPA mitigates confirmation bias. Empirically, UCDPA does not exhibit divergence or oscillation across 1995 runs (3 seeds  $\times$  19 corruptions  $\times$  5 severities  $\times$  7 methods), with no NaN/Inf in any summary file.

**Routing miscalibration.** If thresholds are mis-set, two failure modes arise: (i) too permissive ( $\tau_c$  low,  $\tau_u$  high) amplifies errors via vanilla self-training; (ii) too conservative ( $\tau_c$  high,  $\tau_u$  low) degenerates UCDPA toward COME. The sensitivity sweep (Table 8) characterizes the operating envelope.

## 5.3 Failure Mode Analysis: When UCB Routing Misroutes

**Per-corruption analysis.** Examining per-corruption results (Table 12), UCDPA improves over COME on 18 of 19 corruptions; the single exception is `glass_blur`, where the gain is smaller (+5.01 pp vs. +9.07 pp on `snow`). `glass_blur` produces confidently-wrong samples that the UCB score still routes to the pseudo-label path, muting the gain.

**Error amplification under extreme shift.** On the ResNet-50/ImageNet-C configuration with a corrupted source checkpoint (11.23% source accuracy, near random; Appendix B), UCDPA *underperforms* source accuracy (9.92%). This confirms the theoretical prediction: the error amplification bound relies on  $P(\hat{y} \neq y \mid b > 0) < P(\hat{y} \neq y)$ , which fails when confidence and correctness are uncorrelated. UCDPA is thus *not* a universal remedy—it requires minimum source-model quality below which the dual-path design offers no benefit.

**Mitigation.** For deployment with uncertain source quality: (i) estimate source accuracy on a held-out set (even unlabeled, via confidence-entropy proxies); (ii) use conservative thresholds ( $\tau_c = 0.9$ ,  $\tau_u = 0.1$ ) to default to the COME path; (iii) monitor  $\mathcal{L}_{\text{cal}}$  online—a sudden increase signals routing failure.

## 5.4 Limitations and Threats to Validity

**Limitations.** (i) The UCB threshold depends on four hyperparameters; accuracy varies by up to 2.54 pp across the tested range (Table 8). Adaptive thresholds could improve robustness but introduce additional hyperparameters. (ii) At  $B = 1$ , the calibration regularizer’s batch-mean estimate becomes noisy (ECE rises from 4.27% to 18.95%); per-sample calibration methods could mitigate this. (iii) ResNet-50/ImageNet-C evaluation was inconclusive due to a corrupted source checkpoint; full cross-architecture validation is deferred to future work. (iv) We evaluate on ImageNet-C and CIFAR-100-C; validation on additional benchmarks (e.g., ImageNet-R, domain adaptation datasets) would strengthen the claims.

**Threats to validity.** *Internal:* the improvement over COME could be confounded by implementation differences; we mitigate this by reusing the COME codebase and ablating each component (Table 6). The paired permutation test ( $p < 0.001$ ) and bootstrap CI rule out chance. *External:* results on ImageNet-C with ViT-B/16 may not generalize to all backbones/benchmarks; we report per-corruption and per-severity breakdowns to characterize where the method helps most. *Construct:* ECE/MCE have known binning sensitivity; we use 15 bins following Guo et al. (2017) and results are robust to 10–20 bins. *Statistical:* 3 seeds is standard in TTA work (each ImageNet-C run is 50,000 images per corruption per severity); the bootstrap (10,000 resamples) is the standard non-parametric approach.

## 6 Conclusion

We presented UCDPA (Uncertainty-Calibrated Dual-Path Adaptation), a test-time adaptation method that addresses three limitations of the COME baseline: uniform treatment of samples, no pseudo-label leverage, and no mechanism linking softmax and evidential uncertainty. UCDPA introduces (i) the UCB (Uncertainty-Confidence Balance) score, a differentiable per-sample routing signal that is monotonically increasing in

softmax confidence and monotonically decreasing in Dirichlet uncertainty (Theorem 1); (ii) a dual-path adaptation loss that softly routes each sample between conservative opinion-entropy minimization (for uncertain samples) and pseudo-label cross-entropy (for confident samples), with class-balanced reweighting; (iii) a calibration regularizer that maintains batch-mean consistency between softmax confidence and evidential uncertainty (Proposition 1); and (iv) a feature prototype alignment term that stabilizes the representation under shift.

On ImageNet-C with ViT-B/16, UCDPA improves mean corruption accuracy by +3.07 percentage points over COME (69.20 vs. 66.13,  $p < 0.001$ ), with improved calibration (ECE 3.48 vs. 7.22), improved uncertainty quantification (MI 0.6943 vs. 1.3608), and improved error detection (AUROC 85.22 vs. 84.34). UCDPA also outperforms recent baselines including CoTTA, MEMO, POEM, and FOA (Table 7). Ablations show that UCB routing and class-balanced reweighting provide the largest gains; the calibration regularizer contributes marginally at this scale but serves as a theoretically-motivated safeguard against drift under severe distribution shift, and FPA is an optional auxiliary (disabled by default for efficiency).

**Future work.** Three directions seem most promising. First, **complete cross-backbone validation:** while we have validated UCDPA on ViT-B/16 (LayerNorm) and CIFAR-ResNet-18 (BatchNorm), a full ResNet-50/ImageNet-C evaluation with a valid source checkpoint is needed to close the cross-architecture gap. Second, **online threshold adaptation:** replacing the fixed UCB thresholds with online quantile-based or bandit-based adaptation, which could improve robustness to varying shift severity without introducing instability. Third, **conformal prediction:** leveraging the calibrated uncertainty for distribution-free selective prediction at test time, providing formal coverage guarantees on the abstention set—a natural extension of the error-detection results (Table 3) to a setting with guarantees. We also see potential in extending the calibration regularizer to a per-bin (rather than batch-mean) form, which could provide tighter calibration at the cost of additional computation.

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## A Hyperparameters and Threshold Sensitivity

Table 11 lists all UCDPA hyperparameters. We use the same values across all 19 corruptions and 5 severities (no per-corruption tuning). The threshold sensitivity is reported in Table 8 (main paper); accuracy is stable within 2.54 pp across the tested range, with the best performance at the defaults ( $\tau_c = 0.7$ ,  $\tau_u = 0.3$ ,  $T_c = T_u = 0.1$ ).

Table 11: UCDPA hyperparameters.

| Symbol                  | Description               | Value              |
|-------------------------|---------------------------|--------------------|
| $\tau_c$                | UCB confidence threshold  | 0.7                |
| $\tau_u$                | UCB uncertainty threshold | 0.3                |
| $T_c, T_u$              | UCB temperatures          | 0.1                |
| $\lambda_{\text{come}}$ | COME entropy mixing       | 0.03               |
| $\lambda_{\text{cal}}$  | Calibration weight        | 0.02               |
| $\lambda_{\text{fpa}}$  | FPA weight                | 0 (off)            |
| $\beta$                 | Class-counter EMA         | 0.99               |
| lr                      | Adam LR (LayerNorm)       | $2 \times 10^{-4}$ |
| $B$                     | Batch size                | 128                |

## B Per-Corruption and Cross-Backbone Full Results

Table 12 reports per-corruption accuracy for all methods (severity 3, seed 0). UCDPA outperforms all baselines on *every* corruption. Table 13 reports full cross-backbone, cross-dataset results.

**ResNet-50 negative result.** The ResNet-50 source checkpoint produces 11.23% accuracy (near random), indicating a corrupted or mismatched checkpoint. UCDPA achieves 9.92% (below source), confirming the theoretical prediction (Section 5.3): when the source model is near-random, UCB routing cannot distinguish correct from incorrect pseudo-labels, and the error amplification bound loses its force.

## C Batch Size, Scale-Norm, and Bootstrap Details

**Batch size.** Table 14 reports batch size sensitivity. LayerNorm-only adaptation enables  $B = 1$  TTA; accuracy is maintained within 1.84 pp, though ECE increases at small  $B$  due to noisier batch-mean calibration.

**Scale normalization.** Disabling scale normalization yields essentially identical accuracy ( $\Delta = 0.07$  pp) but increases Mutual Information from 0.69 to 1.04, confirming that scale normalization is essential for the pseudo-label path to meaningfully participate in the dual-path objective.

Table 12: Per-corruption accuracy (ImageNet-C, ViT-B/16, severity 3, seed 0). UCDPA outperforms all baselines on every corruption.

| Corruption        | FOA          | CoTTA        | POEM         | MEMO         | COME         | UCDPA        |
|-------------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Gaussian noise    | 60.41        | 65.61        | 66.85        | 67.55        | 68.72        | 70.80        |
| Shot noise        | 58.03        | 63.69        | 66.28        | 67.17        | 67.17        | 69.83        |
| Impulse noise     | 58.85        | 64.63        | 66.93        | 68.18        | 68.33        | 70.66        |
| Defocus blur      | 51.84        | 58.74        | 63.18        | 65.09        | 63.98        | 66.73        |
| Glass blur        | 37.13        | 42.83        | 54.55        | 56.79        | 54.89        | 60.82        |
| Motion blur       | 58.35        | 64.15        | 66.45        | 68.29        | 67.89        | 70.18        |
| Zoom blur         | 45.06        | 49.32        | 58.69        | 59.14        | 57.70        | 61.70        |
| Snow              | 45.15        | 49.65        | 56.04        | 56.28        | 58.24        | 64.00        |
| Frost             | 34.17        | 38.01        | 44.13        | 41.64        | 48.57        | 56.11        |
| Fog               | 70.89        | 73.10        | 74.61        | 74.76        | 75.27        | 76.24        |
| Brightness        | 72.50        | 74.12        | 75.47        | 75.64        | 76.81        | 78.32        |
| Contrast          | 76.69        | 78.02        | 78.42        | 78.68        | 78.55        | 79.38        |
| Elastic transform | 67.77        | 71.02        | 72.86        | 73.13        | 74.70        | 76.30        |
| Pixelate          | 69.21        | 71.35        | 72.67        | 72.76        | 74.81        | 76.49        |
| JPEG              | 67.99        | 70.65        | 71.88        | 71.92        | 73.04        | 74.76        |
| Speckle noise     | 61.16        | 66.16        | 68.26        | 69.06        | 69.64        | 71.82        |
| Gaussian blur     | 54.51        | 60.67        | 65.29        | 66.91        | 65.92        | 68.61        |
| Spatter           | 64.01        | 67.10        | 68.73        | 68.98        | 70.72        | 73.35        |
| Saturate          | 75.03        | 76.43        | 76.86        | 77.18        | 79.00        | 79.65        |
| <b>Mean</b>       | <b>59.41</b> | <b>63.44</b> | <b>66.75</b> | <b>67.32</b> | <b>68.10</b> | <b>70.83</b> |

**Bootstrap CI methodology.** We report 95% bootstrap CIs for the UCDPA-COME accuracy *difference*  $\Delta$ . For each of 3 seeds, we compute per-corruption  $\delta_{s,c} = \text{acc}_{\text{UCDPA}}(s, c) - \text{acc}_{\text{COME}}(s, c)$ , pool the 57 values (3 seeds  $\times$  19 corruptions), resample 10,000 replicates of size 57 with replacement, and report the 2.5/97.5 percentiles of the mean-difference distribution. The paired permutation test randomly swaps UCDPA/COME labels within each (seed, corruption) pair 10,000 times; the reported  $p$ -value is the fraction of permutations with mean difference at least as large as observed.

## D Positioning: UCDPA vs. Recent TTA Paradigms

Table 15 positions UCDPA relative to recent TTA paradigms. UCDPA achieves high calibration focus at near-baseline compute cost.

Table 13: Cross-backbone, cross-dataset, and recent-baseline results (severity 3, seed 0, mean over 19 corruptions). ResNet-50/ImageNet-C is a *negative* result due to a corrupted source checkpoint (source accuracy 11.23%). AUROC uses the combined score (confidence + evidential uncertainty).

| Backbone / Dataset           | Method       | Acc (%) | ECE (%) | AUROC |
|------------------------------|--------------|---------|---------|-------|
| ViT-B/16, ImageNet-C         | Source       | 59.41   | 4.50    | 0.841 |
|                              | FOA          | 59.41   | 4.50    | 0.841 |
|                              | CoTTA        | 63.44   | 5.97    | 0.844 |
|                              | POEM         | 66.75   | 9.39    | 0.840 |
|                              | MEMO         | 67.32   | 8.77    | 0.838 |
|                              | COME         | 68.10   | 7.23    | 0.848 |
|                              | Tent-COME    | 65.35   | 6.54    | 0.842 |
|                              | <b>UCDPA</b> | 70.83   | 3.27    | 0.854 |
| CIFAR-ResNet-18, CIFAR-100-C | Source       | 62.66   | 8.36    | 0.846 |
|                              | COME         | 62.84   | 8.37    | 0.846 |
|                              | Tent-COME    | 62.94   | 8.45    | 0.847 |
|                              | <b>UCDPA</b> | 63.01   | 8.36    | 0.847 |
| ResNet-50, ImageNet-C (neg.) | Source       | 11.23   | 8.56    | 0.717 |
|                              | COME         | 11.45   | 12.36   | 0.721 |
|                              | <b>UCDPA</b> | 9.92    | 17.78   | 0.684 |

Table 14: Batch size sensitivity (ImageNet-C, ViT-B/16, severity 3, seed 0).

| $B$ | Acc (%) | ECE (%) | AUROC | Runtime (s) |
|-----|---------|---------|-------|-------------|
| 1   | 73.01   | 18.95   | 0.832 | 1162.6      |
| 8   | 74.30   | 13.15   | 0.842 | 343.4       |
| 32  | 73.57   | 7.14    | 0.847 | 300.5       |
| 64  | 72.46   | 4.27    | 0.851 | 291.3       |

Table 15: Positioning of UCDPA relative to recent TTA paradigms.

| Method family               | Signal               | Cost                            | Calib.      |
|-----------------------------|----------------------|---------------------------------|-------------|
| Entropy-min (Tent/EATA/SAR) | Softmax entropy      | $1 \times$                      | Low         |
| Spectral (IMSE)             | Feature stats        | $1 \times$                      | Med         |
| Energy-based (EPOTTA)       | Energy               | $1.1 \times$                    | High        |
| Particle-based              | Multi-hyp.           | $K \times$                      | High        |
| Evidential (COME)           | Dirichlet ent.       | $1.02 \times$                   | Med         |
| <b>UCDPA</b>                | <b>Evid. routing</b> | <b><math>1.01 \times</math></b> | <b>High</b> |

**Algorithm 1** UCDPA: Uncertainty-Calibrated Dual-Path Adaptation

**Require:** Source model  $\theta_0$ ; data stream  $\{\mathbf{x}_t\}_{t=1}^T$ ; hyperparameters  $\tau_c, \tau_u, T_c, T_u, \lambda_{\text{come}}, \lambda_{\text{cal}}, \lambda_{\text{fpa}}, m, \beta, \text{lr}$ ; batch size  $B$ .

- 1: Initialize LayerNorm parameters  $\theta \leftarrow \theta_0$ ; freeze all non-LayerNorm parameters.
- 2: Initialize EMA class counter  $n_c \leftarrow 0$  for all  $c$ ; prototype bank  $\mu_c \leftarrow \mathbf{0}$  for all  $c$ .
- 3: **for**  $t = 1, 2, \dots, T$  **do**
- 4:   Receive batch  $\mathcal{B}_t = \{\mathbf{x}_i\}_{i=1}^B$ .
- 5:   Compute logits  $\mathbf{z}_i = f_\theta(\mathbf{x}_i)$ , softmax  $p_i = \text{softmax}(\mathbf{z}_i)$ , confidence  $c_i = \max_k p_{ik}$ .
- 6:   Evidential computation (COME).
- 7:    $\tilde{\mathbf{z}}_i \leftarrow \mathbf{z}_i / (\|\mathbf{z}_i\|_2 + 1)$ ;  $e_{ik} \leftarrow \exp(\text{ReLU}(\tilde{z}_{ik})) - 1$ .
- 8:    $\alpha_{ik} \leftarrow e_{ik} + 1$ ;  $S_i \leftarrow \sum_k \alpha_{ik}$ ;  $b_{ik} \leftarrow e_{ik}/S_i$ ;  $u_i \leftarrow K / S_i$ .
- 9:   UCB routing score (thresholds  $\tau_c, \tau_u$ ; temperatures  $T_c, T_u$ ).
- 10:    $\hat{c}_i \leftarrow \text{sg}(c_i)$ ,  $\hat{u}_i \leftarrow \text{sg}(u_i)$    // detach for routing.
- 11:    $b_i \leftarrow \sigma((\hat{c}_i - \tau_c)/T_c) \cdot \sigma((1 - \hat{u}_i - \tau_u)/T_u)$ .
- 12:   // Pseudo-labels and class-balanced weights (with safeguards).
- 13:    $y_i^* \leftarrow \arg \max_k p_{ik}$ .
- 14:   Update EMA counter:  $n_c \leftarrow \beta n_c + (1 - \beta) \sum_i \mathbf{1}\{y_i^* = c\}$ ;  $n_c \leftarrow \max(n_c, 0.01)$ .
- 15:    $w_{\text{cb},i} \leftarrow \text{clip}(K \cdot \tilde{w}_{y_i^*}, 0.5, 2.0)$  with  $\tilde{w}_c \propto 1/(n_c + \epsilon)$ .
- 16:   // Path 1: conservative entropy.
- 17:    $\mathcal{L}_{\text{come},i} \leftarrow (1 - \lambda_{\text{come}}) H(p_i) + \lambda_{\text{come}} H_{\text{opinion}}(\mathbf{o}_i)$ .
- 18:   // Path 2: pseudo-label cross-entropy, scale-normalized (batch-mean detached).
- 19:    $\tilde{\mathcal{L}}_{\text{pl},i} \leftarrow \mathcal{L}_{\text{pl},i} \cdot \text{sg}(\mathbb{E}_{\mathcal{B}}[\mathcal{L}_{\text{come}}]) / (\text{sg}(\mathbb{E}_{\mathcal{B}}[\mathcal{L}_{\text{pl}}]) + \epsilon)$ .
- 20:   // Dual-path loss (with detached UCB routing  $b_i$ ).
- 21:    $\mathcal{L}_{\text{dual}} \leftarrow \frac{1}{B} \sum_i [\text{sg}(1 - b_i) w_{\text{cb},i} \mathcal{L}_{\text{come},i} + \text{sg}(b_i) w_{\text{cb},i} \tilde{\mathcal{L}}_{\text{pl},i}]$ .
- 22:   // Calibration regularizer ( $c_i, u_i$  **not** detached — gradient flows to  $\theta$ ).
- 23:    $\mathcal{L}_{\text{cal}} \leftarrow |\text{mean}(c_i) - \text{mean}(1 - u_i)|$ .
- 24:   // Feature prototype alignment.
- 25:   Extract penultimate features  $\mathbf{f}_i$ ; update prototypes  $\mu_c \leftarrow m \mu_c + (1 - m) \text{mean}_{i: y_i^* = c} \mathbf{f}_i$ .
- 26:    $\mathcal{L}_{\text{fpa}} \leftarrow \frac{1}{B} \sum_i \text{CE}(\text{sim}(\mathbf{f}_i, \mu_{y_i^*}) / T_{\text{fpa}}, y_i^*)$ .
- 27:   // Total loss and update.
- 28:    $\mathcal{L} \leftarrow \mathcal{L}_{\text{dual}} + \lambda_{\text{cal}} \mathcal{L}_{\text{cal}} + \lambda_{\text{fpa}} \mathcal{L}_{\text{fpa}}$ .
- 29:   Update LayerNorm parameters  $\theta$  via Adam on  $\mathcal{L}$  with learning rate  $\text{lr}$ .
- 30: **end for**